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The neuroprotective effects of baobab and black seed on the rat hippocampus exposed to a 900-MHz electromagnetic field

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ABSTRACT

This study investigated the potential effects on the hippocampus of electromagnetic fields (EMFs) disseminated by mobile phones and the roles of baobab (*Adansonia digitata*) (AD) and black seed (*Nigella sativa*) (BS) in mitigating these. Fifty-six male, 12-week-old *Wistar albino* rats were divided into eight groups of seven animals each. No EMF exposure was applied to the control, AD or BS groups, while the rats in the Sham group were placed in an EMF system with no exposure. A 900-MHz EMF was applied to the EMF+AD, EMF+BS, EMF+AD+BS and EMF groups for 1 hour a day for 28 days. Pyramidal neurons in the hippocampus were subsequently counted using the optical fractionator technique, one of the unbiased stereological methods. Tissue sections were also evaluated histopathologically under light and electron microscopy. The activities of the enzymes catalase (CAT) and superoxide dismutase (SOD) were also determined in blood serum samples. Analysis of the stereological data revealed no statistically significant differences between the EMF and control or sham groups in terms of pyramidal neuron numbers ($p>0.05$). However, stereological examination revealed a crucial difference in the entire hippocampus between the control group and the AD ($p<0.01$) and BS ($p<0.05$) groups. Moreover, exposure to 900-MHz EMF produced adverse changes in the structures of neurons at histopathological analysis. Qualitative examinations suggest that a combination of herbal products such as AD and BS exerts a protective effect against such EMF side-effects.

1. Introduction

Mobile phones have become an integral part of modern communications (Ebrahim et al., 2016; Linhares, 2017). Numerous epidemiological studies have confirmed the risk to human health posed by cell phone use (Deniz and Kaplan, 2022; Elamin et al., 2022). Since these tools are used close to the head, the brain may be particularly affected following exposure to electromagnetic fields (EMFs) (Erdem Koç et al., 2016; Deniz et al., 2017). The side-effects of EMF exposure, which generates a thermal effect in the targeted tissues, are still controversial. The effects of EMF exposure on human health are not yet fully understood due to a scarcity of studies on the subject and the complexity of the body. Some previous studies demonstrating the detrimental effects of EMF exposure on organisms have mainly focused on the nervous system, especially the hippocampus, a region known to regulate learning and memory (Kivrak et al., 2017; Deniz and Kaplan, 2022; Elamin et al., 2022).

Oxidative stress can lead to cell membrane damage by altering the structure of proteins and through the peroxidation of lipids (Pizzino et al., 2017). Numerous studies have demonstrated that the brain is sensitive to oxidative stress because of its high metabolic rate and oxygen requirement (Wang and Michaelis, 2010; Liu et al., 2017). The brain possesses different mechanisms for responding to oxidative stress by producing antioxidants, either generated naturally in situ (endogenous antioxidants) or supplied externally through diet (Pizzino et al., 2017). Antioxidant systems inhibit neoplastic processes and neutralize free radicals in cells and tissues by developing an enzymatic defence system (Lü et al., 2010; Chaudhary et al., 2023).

The pulp of baobab (*Adansonia digitata*, AD), which is frequently employed in African countries, is rich in minerals, vitamins (such as vitamin C) and organic and essential amino acids and flavonoids. It therefore represents an effective antioxidant, with other properties such as anti-inflammatory, analgesic and antipyretic activities. It is also an antimicrobial, antiviral and antidiarrheal agent with a hepatoprotective

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role (Kabore et al., 2011).

Black seed (*Nigella sativa*) (BS) is a medicinal plant used widely across the world and favoured in several indigenous medicinal systems (Mohamadin et al., 2010). Numerous active compounds have been derived from distinctive dark seed mixtures, the most important being thymoquinone (TQ), p-cymene, carvacrol, 4-terpineol, α-pinene and thymol (Beheshti et al., 2016). *Nigella sativa* is used as an anticancer agent and is reported to possess anti-diabetic, diuretic, anti-inflammatory, antimicrobial, anticonvulsant, anti-nociceptive, anti-anxiety, antioxidant, nephroprotective and immunomodulatory activities (Gholamnezhad et al., 2014; Abdel-Daim and Ghazy, 2015).

In light of this information, this study aimed to examine the potential effects of AD and BS on the rat hippocampus following exposure to 900-MHz EMF radiation. So far as we are aware, there have been no stereological and electron microscopic studies of the neuroprotective effects of AD and BS on the hippocampus following EMF exposure. The potential neuroprotective effects on adult hippocampal neurogenesis after EMF exposure of these two agents are therefore still unknown. We believe that this study using quantitative methods and histopathological analysis will shed new light on the subject.

2. Materials and methods

2.1. Experimental design

Approval for the study was received from the Ondokuz Mayis University local animal experiments ethics committee (No. 2018/17 dated 30.03.2018). All animal procedures were carried out in compliance with the EU Directive 2010/63/EU for animal research, the U.K. Animals (Scientific Procedures) Act of 1986, and the ARRIVE standards. These rules were scrupulously followed when performing the experiment. Fifty-six male Wistar albino rats were divided into eight groups of seven rats each, with one animal from each group being selected and used for electron microscopic examinations.

1. **Control group:** No experimental procedure was performed on this group for the 28-day study period.
2. **Sham group:** This group was placed in a special chamber for 1 hour a day over a period of 28 days, similarly to the EMF groups. However, this group was not exposed to EMF radiation.
3. **EMF group:** This group was exposed to 900 MHz EMF for 1 hour a day for 28 days using a specific EMF system (Ulubay et al., 2015).
4. **EMF+AD group:** This group received AD (300 mg/kg/day) by oral gavage and was exposed to EMF for 28 days.
5. **EMF+BS group:** This group received BS (400 mg/kg/day) by oral gavage and was exposed to EMF for 28 days.
6. **EMF+AD+BS group:** This group received AD (300 mg/kg/day) and BS (400 mg/kg/day) oral gavage and was exposed to EMF for 28 days.
7. **AD group:** This group received AD (300 mg/kg/day) by oral gavage for 28 days.
8. **BS group:** This group received BS (400 mg/kg/day) by oral gavage for 28 days.

The absorbed electromagnetic energy in tissue is quantified using SAR (W/kg) because of EMF power absorption in the rat body, or else using the maximum permissible exposure value of EM radiation power density (mW/cm) (Martens et al., 1995). In the present study, we used the electric fields unit as the exposure value parameter (mV/m). In addition, EMF exposure during the experiments was performed in an isolated room in order to reduce unwanted exposure from external sources. The 12-week-old rats (weighing 300±50 g) were kept at 22–25°C and humidity of 50–60% in a 12-h light:dark cycle. After 28 days, rats were anaesthetized by the intraperitoneal administration of 5 mg/kg ketamine (Ketlar®, Pfizer, Istanbul, Turkey) and 2 mg/kg xylazine (Xylazinbio®, Intermed pharmacy store, Ankara, Turkey). All the animals were then sacrificed, after which the brains were removed.

2.2. Light and electron microscopic tissue procedures

Following sacrifice, the right cerebral hemispheres were placed in 10% formalin for 10 days for stereological analysis. Paraffin (Merck, Darmstadt, Germany) was then used to embed the brain tissues following routine histological procedures. Tissue sections 20 μm in thickness (1/6 sample) were then collected using a microtome (Leica RM2245; Leica Instruments, Nussloch, Germany) and stained with 0.1 g cresyl violet in 100 ml of distilled water at room temperature. The sections were then passed through graded 70–100 alcohol series (5–10 min for each), followed by xylene. A cover glass with Entellan™ was used for mounting the sections, which were finally evaluated stereologically using the optical fractionator technique (Deniz and Kaplan, 2022; Elamin et al., 2022).

One brain from each group was processed for resin embedding. The right cerebral hemispheres from all groups were fixed in a 4% paraformaldehyde and 4% glutaraldehyde mixture for seven days. At the end of that period, hippocampal samples 1 mm^3 in volume were washed with 0.1 M phosphate buffer saline (PBS), which has a pH of 7.4. The specimens were subsequently placed into propylene oxide, dehydrated in an increasing acetone series, washed in PBS, post-fixed in 1% osmium tetroxide, and finally placed into a 50% propylene oxide+50% Araldite combination. The final procedure to prepare tissues for electron microscopy involved overnight embedding in a mixture of Araldite placed in the silicon embedding mould. The mould containing the samples was placed in an oven at 70° C for polymerization for two days. Semi-thin (0.5 μm) and ultra-thin (70–80 nm) sections were taken from the polymerized resin blocks using an ultra-microtome with a diamond blade (Leica Ultracut UCT; Leica Microsystems GmbH, Wetzlar, Germany). One percent toluidine blue was then used to stain the semi-thin sections. In addition, the ultra-thin sections were placed onto a metal grid and stained with the uranyl acetate and lead citrate (Leica, EM AC20) (Şimşek et al., 2009; Kaplan et al., 2013). After these routine procedures, the ultra-thin sections were evaluated to observe any changes in the ultrastructure of the hippocampus under a transmission electron microscope (JEOL-JSM- 7001 F). The semi-thin sections were observed under a light microscope for histopathological examinations.

2.3. Stereological analysis

Stereological methods are used to provide quantitative and unbiased estimates yielding three-dimensional data from two-dimensional sections (Yurt et al., 2018). These methods allow biological structures to be examined reliably and impartially as a whole entity. The optical fractionator technique makes it possible to calculate the total numbers of cell or particles in thick sections. A stereological probe, optical disector, and systematic random sampling are applied together when using this method (Gundersen and Jensen, 1987; West et al., 1991). This technique represents a combination of the optical disector and fractionator methods (Gundersen, 1986; West et al., 1991). It includes countable particles using optical disectors in a systematic sample representing an known area which is due to be examined. This method is not affected by the size and orientation of the particles or by shrinkage or expansion in the structure during histological tissue tracking procedures. It is also one of the most effective stereological counting techniques (West et al., 1991). The first step involves the collection of sections according to systematic random sampling fractionation. This is known as the section-sampling fraction (ssf). The ssf in the present study was 1/12. The second step involves counting the particles using an unbiased counting frame on a regular grid consisting of x and y positions for each sampled area. The is known as the area-sampling fraction (asf). The asf in the present study was 3600 μm^2/14,400 μm^2 (counting frame area/step area). The upper and lower guard zones were set at 3 μm. The counting unit of the present study was the neuronal nucleus.

The third step includes the section thickness fraction (stf), comprising the disector and the heights of the section thickness. These values were 12 μm for the disector and 20.9 μm for the section thickness in the present study, respectively. Following these steps, the total cell number in a tissue can be estimated using the formula

$$N = \sum Q^{-} \cdot \frac{1}{ssf} \cdot \frac{1}{asf} \cdot \frac{1}{tsf}$$

where ΣQ- symbolizes the disector pyramidal cell number counted in all hippocampal sections and "tsf" represents the part of the examined section in the z-axis (Elamin et al., 2022). The coefficient of variation (CV) and estimation of the coefficient of error (CE) were estimated by means of a pilot study (Gundersen, 1986; Kaplan et al., 2013) (Table 1).

2.4. Biochemical analysis

Intracardiac blood was drawn from the rat hearts while they were under anaesthesia and prior to sacrifice. These were then placed in blood collection tubes before centrifugation at 4° C (Hermle Labor Technik GmbH, Germany) for 15 minutes at 2000 pm. The resulting serum samples were kept at −80°C until being used for biochemical analysis. The CAT and SOD enzyme activities were analyzed by measuring absorbance values.

2.5. Statistical analysis

IBM SPSS 21.0 version software was used for statistical analyses (SPSS Inc., Chicago, IL, USA). The data were expressed as mean ±standard deviation. After the Shapiro-Wilk normality test had been applied, the One-Way ANOVA test was used to assess differences between the groups, and the Tukey test was applied as post-hoc. p values <0.05 were regarded as statistically significant.

3. Results

3.1. Stereological results

3.1.1. Total pyramidal cell numbers in the CA regions

Pyramidal neuron numbers were assessed for the CA1+CA2 and CA3 areas, as well as for the entire hippocampus. Stereological data from the CA1+CA2 hippocampal regions revealed that the mean number of neurons in the EMF group was much lower than in the EMF+AD+BS group ($p<0.05$). Additionally, the mean number of neurons in the control group differed from those in the BS, AD and EMF+AD+BS groups ($p<0.01$) (Fig. 1**A**).

Pyramidal neurons in the CA3 region decreased significantly in the EMF group compared to the EMF+AD+BS group ($p<0.05$). In addition, the mean total number of neurons in the control group was also considerably lower than in the EMF+AD+BS group ($p<0.05$). A significant increase was observed in the mean number of pyramidal neurons in the EMF+AD+BS group compared with the EMF+AD ($p<0.05$) and EMF+BS ($p<0.01$) groups. Similarly, the AD group exhibited a substantial rise in the mean number of pyramidal neurons compared to the EMF+AD group ($p<0.01$) (Fig. 1**B**).

Table 1
The mean CE and CV values of pyramidal cell numbers estimated in the CA regions.

Group		CA1+CA2	CA3	All CA regions
Cont	CE	0.04	0.02	0.02
	CV	0.03	0.03	0.03
Sham	CE	0.04	0.02	0.02
	CV	0.06	0.08	0.07
EMF	CE	0.05	0.05	0.02
	CV	0.05	0.04	0.06
EMF+AD	CE	0.02	0.02	0.02
	CV	0.08	0.07	0.08
EMF+BS	CE	0.02	0.03	0.02
	CV	0.04	0.06	0.05
EMF+AD+BS	CE	0.03	0.02	0.03
	CV	0.03	0.02	0.02
AD	CE	0.03	0.025	0.03
	CV	0.03	0.02	0.03
BS	CE	0.03	0.02	0.03
	CV	0.01	0.02	0.01

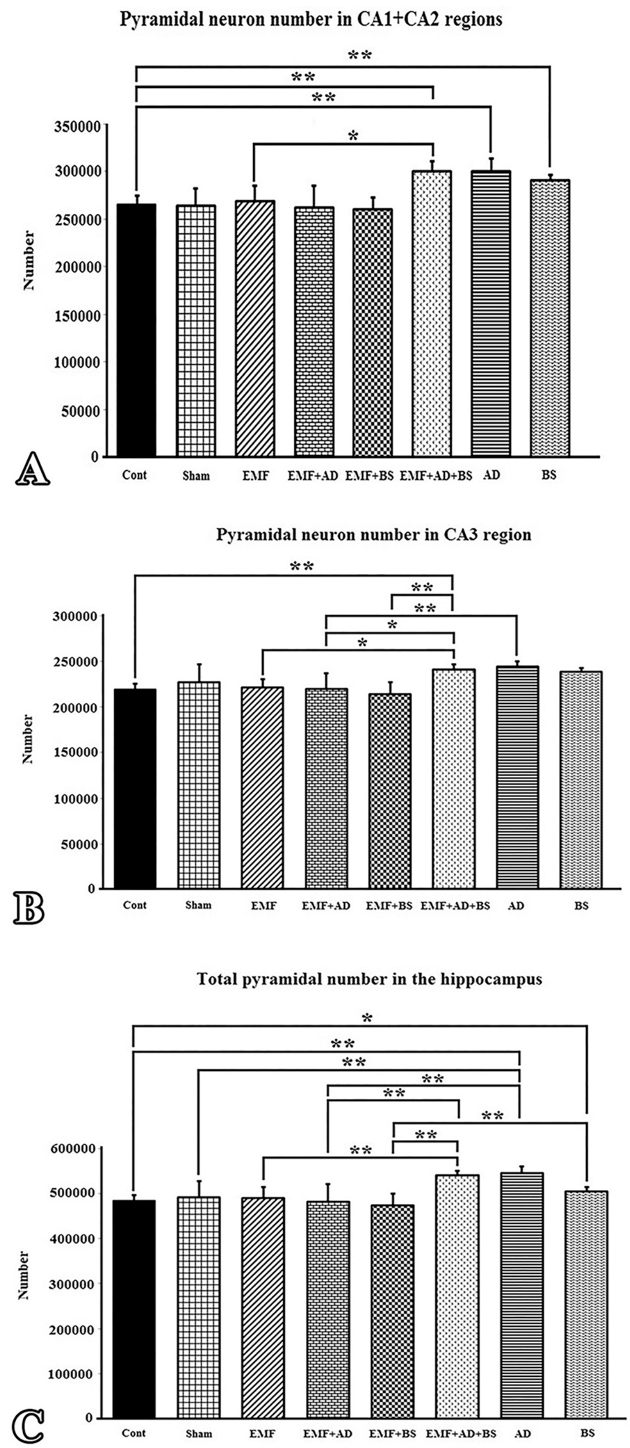


Fig. 1. Differences in the number of pyramidal neurons in the CA1+CA2, CA3 areas, and the total hippocampus, following exposure to EMF for 1 hour. Significant differences at the $p<0.05$ level are denoted by (*), and those at the $p<0.01$ level by (**).

The mean total pyramidal neuron numbers in the entire CA region exhibited a crucial decrease in the EMF group compared to the EMF+AD+BS and AD groups ($p<0.01$). However, the total number of pyramidal neurons in the control group CA region differed significantly from those in the AD ($p<0.01$) and BS ($p<0.05$) groups. Moreover, total pyramidal neuron numbers in the CA region of the Sham group differed significantly from those in the AD group ($p<0.01$). Additionally, a significant increase was observed in mean total pyramidal neuron numbers in the EMF+AD+BS group compared with the EMF+AD and EMF+BS ($p<0.01$) groups. (Fig. 1**C**).

3.2. Histopathological investigations of the hippocampal sections under the light microscopy using resin sections

Light microscopic sections from the CA region exhibited substantial numbers of healthy pyramidal neurons with clear boundaries in the control, Sham, AD and BS groups. The neuron borders and cell nuclei of the EMF+AD and EMF+BS groups were clearly delineated, and the structure of the neurons was typically well preserved at morphological analysis. These groups possessed a higher density of healthy neurons than the EMF group. Cells with darkly coloured nuclei and cytoplasm were more prevalent in the EMF group. In addition, the EMF group possessed more indistinct boundaries and degenerated (dark-stained)

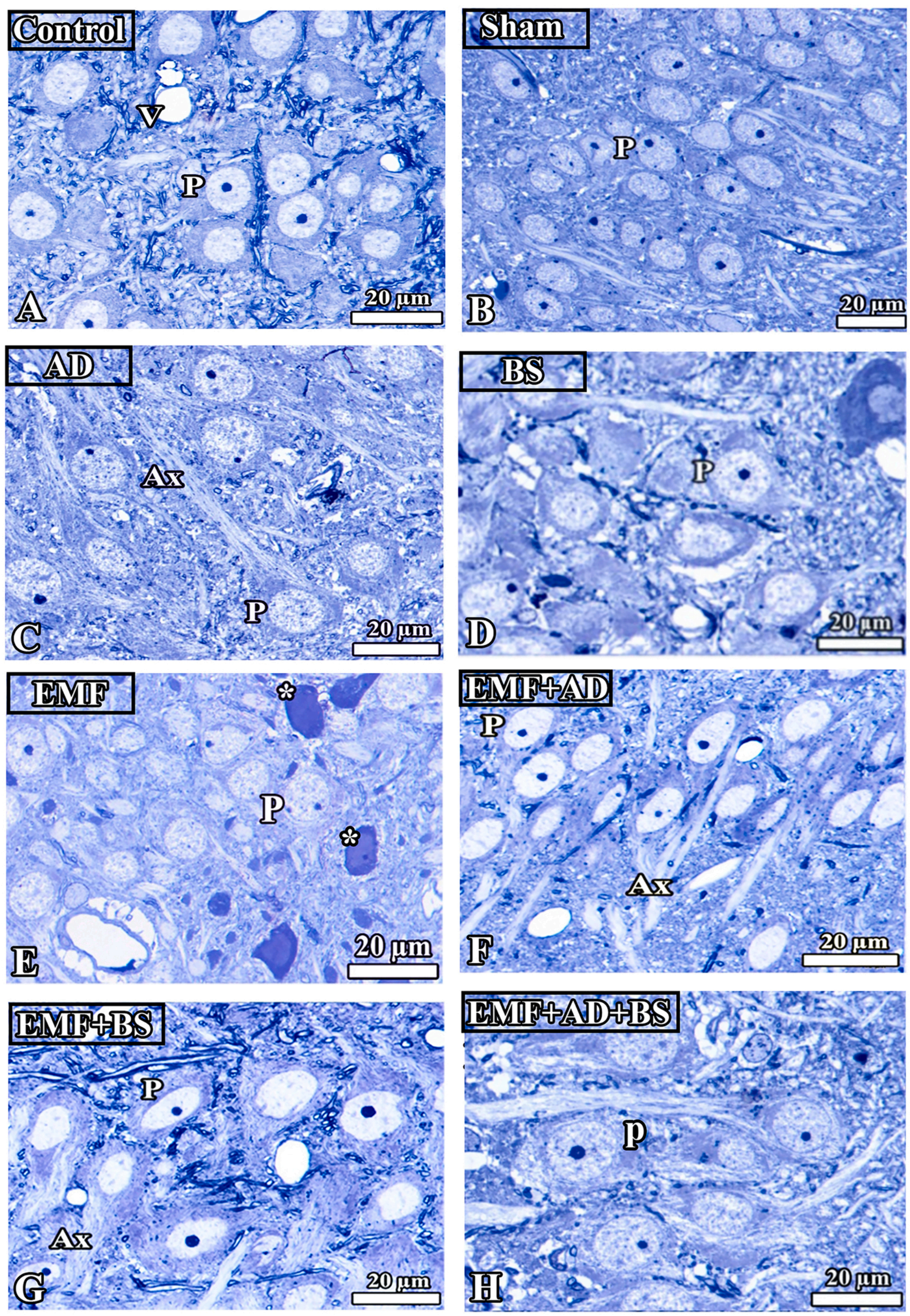


Fig. 2. Images of toluidine blue-stained semi-thin sections taken from the CA area of the hippocampus after being exposed to EMF for 1 hour. (*) represents degeneration neurons and (P) shows healthy pyramidal neurons. (v): vessel; (Ax): Pyramidal cell axon.

neurons than the EMF+AD and EMF+BS groups. Pyramidal neurons in the BS group were often regular in shape and exhibited a healthy appearance. The borders of the neuronal perikarya could be clearly delineated clearly. This group exhibited a normal appearance in terms of the general structure of pyramidal neurons (Fig. 2).

3.3. Histopathological investigations of the hippocampal sections under electron microscopy

Electron microscopic images from the control and Sham groups showed that the pyramidal cells were oval in shape and healthy, with a

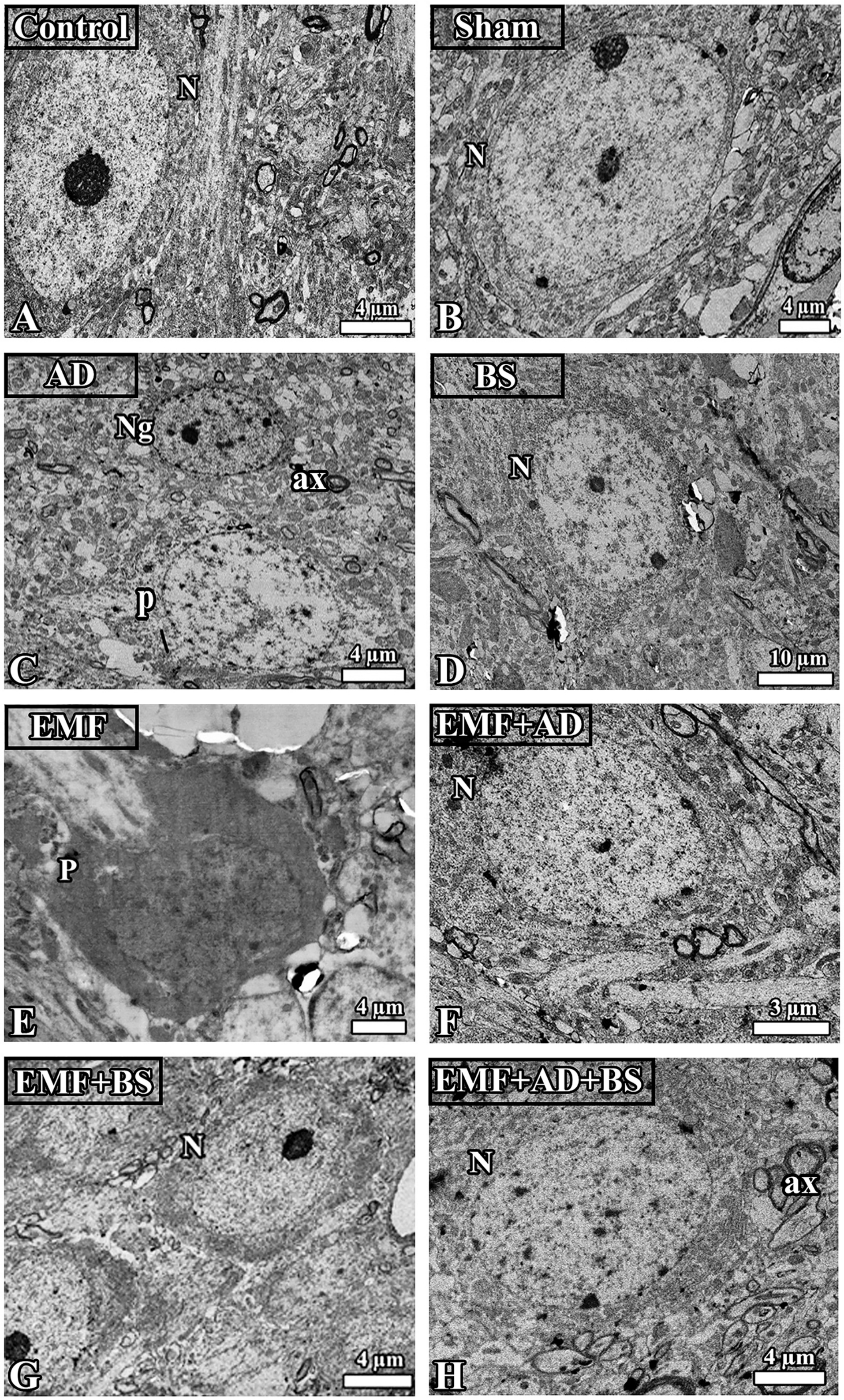


Fig. 3. Electron microscopic images from the CA region of the hippocampus following 1-hour exposure to EMF. (N) indicated the pyramidal cell nucleus and (P) a degenerated pyramidal neuron. (Ng): Neuroglia.

condensed cytoplasm. Electron microscopic images from the AD and BS groups revealed that the healthy pyramidal cells were oval-shaped with a euchromatic nucleus. Some neuroglia and myelinated axons were also observed in the AD group, and these exhibited a normal morphology. Images from the EMF group showed that numerous pyramidal cells in the hippocampus were deformed, with most being dark-stained. Many degenerated pyramidal neurons with shrunken perikarya could be seen, meaning that these had no function and were headed to death. Additionally, images from the EMF+AD and EMF+BS groups revealed healthy pyramidal cells with oval-shaped nuclei and pronounced nucleoli with slightly dark-stained cytoplasm. Myelinated axons could also be seen in the EMF+AD group and these axons were also healthy in appearance. Additionally, images from the EMF+AD+BS group exhibited pyramidal cells with a normal morphology and clearly distinct borders, together with some accumulation of myelinated axons around them (Fig. 3).

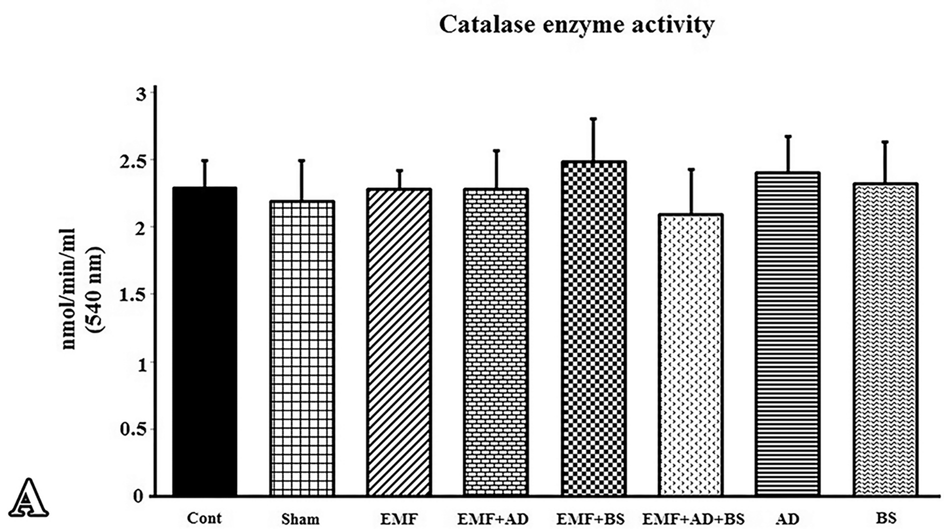
3.4. Biochemical findings

3.4.1. CAT enzyme activity

Blood serum samples from all the study groups were examined for CAT enzyme activity. The results are shown in Fig. 4**A**. No statistically significant differences were observed among the groups ($p>0.05$).

3.4.2. SOD enzyme activity

Blood serum samples collected from all the study groups were also examined for SOD enzyme activity. The results are shown in Fig. 4**B**. No significant difference in SOD activity levels was observed between the EMF group and the EMF+AD, EMF+BS and EMF+AD+BS groups ($p>0.05$). No significant difference in SOD activity was also determined among the control, Sham and EMF groups ($p>0.05$). Enzyme activity was lower in the control and Sham groups than in the EMF group, although these differences were not statistically significant. However, a significant difference in the activity of this enzyme was observed between the EMF+AD group and the BS group ($p<0.01$). There was no significant difference between the EMF+AD and AD groups ($p>0.05$).



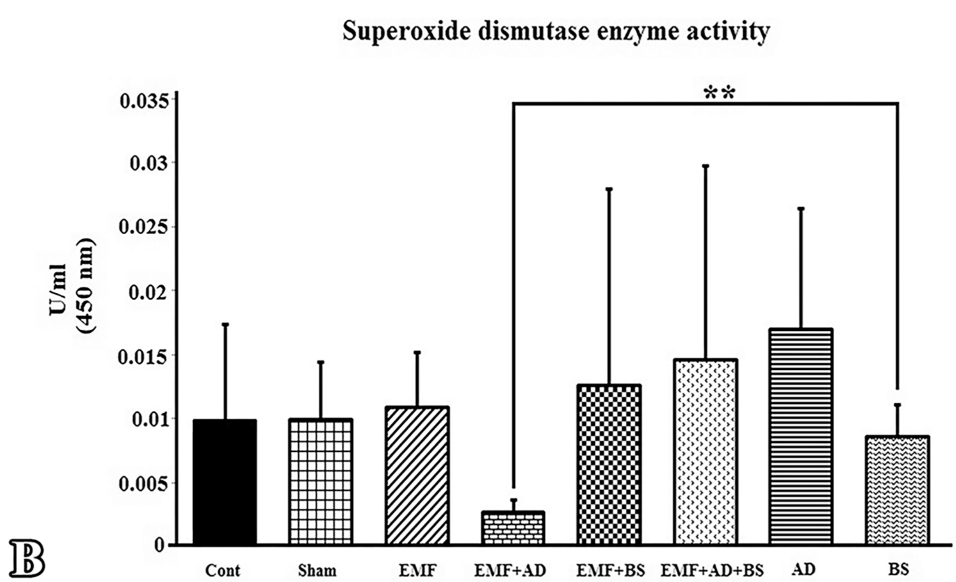


Fig. 4. **(A)** Mean values for each group's CAT enzyme activity. No significantly different results were found. The study groups' mean SOD enzyme activity values are shown in **(B).** Differences that are statistically significant at $p<0.01$ are denoted by (**).

The difference observed between the BS and EMF+AD groups may indicate that BS is a more effective antioxidant than AD. The absence of any significant difference between the BS and AD groups may be due to statistical deviation in the AD group. More advanced experimental studies are now needed to elucidate the antioxidant effects of BS and AD.

3.5. Total phenolic content and antioxidant power findings

This study also investigated the total phenolic content, the capacity of the antioxidants, and the free radical scavenging effects of AD and BS. The findings are shown in Table 2.

4. Discussion

Stereological methods yield rapid, efficient, high-quality and unbiased results (Altun et al., 2017). Our stereological analysis revealed no significant decrease between the EMF group and the control group in terms of mean total pyramidal neuron numbers in the CA1+CA2 or CA3 areas. This may be attributable to the insufficient EMF exposure time. Further studies on this subject are therefore needed. In addition, no change was observed in the mean total pyramidal neuron numbers in the Sham group compared to the control group. However, mean total pyramidal cell numbers in the CA regions were significantly higher in the AD and BS groups than in the control group. This may indicate that a herbal product containing a high level of antioxidants can stimulate neurogenesis in this region. However, this idea now needs to be confirmed through immunohistochemical analysis, and newly generated neurons in this region must also be shown using special staining techniques.

Previous studies have reported that the pyramidal neurons in the hippocampal CA region can be damaged by one hour of exposure each day for 28 days (Şahin et al., 2015; Kim et al., 2016). The number of reports of the side-effects of EMR emitted by mobile phones is also increasing (Ibrahim and Gharib, 2010). Studies have shown that EMF exposure leads to an increase in brain cancer rates and reduces the numbers of neurons in the child hippocampus, and also significantly affects cognitive functions (Merzenich et al., 2008; Södergvist et al., 2011).

Some recently published studies using stereological analyses have shown that experimental groups exposed to 900-MHz possessed fewer pyramidal neurons in the CA region compared to other non-exposed groups (Altunkaynak et al., 2016; Kivrak et al., 2017). In the present investigation, pyramidal neurons were counted in each of the CA regions using the optical fractionator method. The numerical data obtained through this technique represent the total number of pyramidal neurons in the hippocampal regions CA1+CA2 and CA3, as well as in the whole hippocampus. One of the pioneer studies of EMF in this context, by Odaci et al. (2008), suggested that exposure to 900 MHz adversely affects the numbers of granule cells in the DG, as well as its morphology, during its development. A previous similar study demonstrated that 900 MHz EMF exposure caused neuronal damage in the CA region (Bas et al., 2009), Another study also noted a crucial decrease in total pyramidal neuron numbers in the CA region of rats exposed to 900 MHz EMF for 60 minutes a day for three weeks compared to a control group (Erdem Koç et al., 2016).

Table 2
The total phenolic content, the capacity of antioxidants and the free radical scavenging effect of AD and BS.

Antioxidant	Total phenolic content, mg / kg, gallic acid equivalent (GAE)	FRAP, mmol/g Trolox equivalent	DPPH free radical removal of mmol / g, Trolox equivalent
AD	14812.5	558.66	102.56
BS	4906.25	54.03	25.02

As described above, some of the stereological results from this study are inconsistent with those from earlier investigations (Şahin et al., 2015; Erdem Koç et al., 2016). However, our findings are broadly consistent with those of Rağbetli et al. (2009). Those authors detected no statistically significant differences in the numbers of pyramidal neurons in the rat hippocampus between a control group and an EMF-exposed group. The stereological data from the present study also revealed no side-effects of EMF radiation in terms of total neuron numbers. However, this was not confirmed by the histopathological evaluation, and this discrepancy should be investigated in further studies.

Several studies have investigated the biological activities of AD and have reported promising outcomes (Abiona et al., 2015; Adegoke et al., 2015). However, our search of the scientific literature revealed no studies of the potential effects of AD on the brain exposed to EMF. The stereological results of this study demonstrated a rise in the total number of pyramidal neurons in the CA region of the hippocampus in the AD group compared to the control and Sham groups. It may be concluded that AD protects the pyramidal neurons in the hippocampus against free radicals, which cause severe cell damage in neural cells. The present study is therefore original from that perspective as the first to focus on the effect of AD on the hippocampus exposed to EMF radiation. Further studies involving different EMF exposure times are now needed.

Analyses of the neuron population in the CA region revealed a substantial increase in the mean total number of pyramidal neurons in the BS group compared to the control group. A previous study demonstrated that the protective effect of BS is associated with its powerful antioxidant, antifibrotic and anti-inflammatory effects and its ability to prevent the formation of free oxygen radicals (Erboga et al., 2015). From that perspective, BS has been shown to exert beneficial effects against neurodegeneration in the rat hippocampus following chronic toluene exposure (Kanter, 2008). However, further studies are now required to reveal the potential protective effects of BS against the EMF in the brain. Studies involving different EMF exposure times are particularly needed.

In terms of histopathological observations, Şahin et al. (2015) reported that EMF alters the morphology of the hippocampus. Their histological findings revealed numerous abnormal dark-stained cells and shrinkage of the perikaryon in the hippocampus in an EMF group compared to a control group. Keleş et al. (2018) also revealed pronounced impairment of pyramidal cells in the CA1, CA2 and CA3 regions and damaged granular cells in the DG region. The present study also observed numerous dark-stained cells under light microscopy in the EMF group. Salford et al. (2003) observed that rats exposed to EMF showed signs of damage to nerve cells in the cortex and hippocampus, including the existence of dark-stained neurons and shrinkage of internal cell components. Additionally, Srivastava and Taufiq's (2018) electron microscopic findings revealed single degenerated cells with dark cytoplasm and nuclei, disordered mitochondrial cristae, and vacuolated mitochondria in different regions of brains from EMF-exposed rats. Faridi and Khan (2013) also observed shrinkage in pyramidal cells due to the deleterious effects of EMF, which may lead to programmed cell death. These morphological changes may be due to decreased nuclear activity.

The histopathological changes in the hippocampus observed under light and electron microscopy in the above studies are compatible with the results of the present research. In contrast to the control and Sham groups, pyramidal neurons in the EMF-exposed group exhibited morphological variation. These findings revealed that dark- or dark blue-stained neurons were damaged, these being more numerous in the EMF group than in the control and Sham groups. The study results suggest that the presence of numerous dark-stained neurons in the EMF group is attributable to the condensation of heterochromatin in the degenerated neurons. Additionally, histopathological changes observed under light and electron microscopy revealed more normal neurons in the EMF+AD, EMF+BS and EMF+AD+BS groups than in the EMF group, since distinct cell and nuclear boundaries were a major characteristic of the those three groups. Moreover, histopathological examination showed that the AD and BS groups possessed healthy and oval-shaped pyramidal cells with a euchromatic nucleus.

Biochemical analysis was also performed on serum blood samples. CAT and SOD enzyme activities in the EMF group were not significantly different to those in the control and Sham groups. This indicates that the level of free radical formation was not exceptionally high. In that context, Altun et al. (2017) reported that blood serum samples from their EMF group exhibited higher levels of SOD and CAT activity in comparison to control and Sham groups. The reduction endogenous antioxidant activity and the promotion of the reactive oxygen species' production, which result cell damage, are both adverse effects of EMF radiation. This can be explained in terms of a rise in free radicals due to EMF exposure. In the present study, CAT and SOD activities had no statistically significant effect on free radical levels. This can be explained in terms of the insufficient duration of EMF exposure. Further studies are now needed from that perspective.

In the light of our study findings, and especially the histopathological outcomes, we conclude that BS and AD can be used as natural herbal agents in order to reduce the cellular effects of radiation-related neurological impairment, particularly when used in a combination, but also when applied separately. However, further research is still required to fully understand how BS and AD affect the brain.

5. Conclusion

We suggest that using appropriate quantities of natural antioxidants in combination with foodstuffs can inhibit or reduce the harmful effects of EMF radiation on the neurons of the brain. The human population, and especially children, should also be protected against exposure to radiation, especially that emitted from mobile phones. To the best of our knowledge, no prior study has demonstrated the effect of AD and BS in the EMF exposed rat hippocampus. Further studies focusing on the effect mechanism of antioxidants, especially AD and BS, that may represent novel protective substances against the side-effects of EMF radiation in the hippocampus, are now needed. Research involving different methods, durations and doses is therefore required.

Funding information

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Ethics approval statement

The Ondokuz Mayis University Local Ethics Committee of Animal Experiments gave its approval for the relevant study (No. 2018/17 dated 30.03.2018).

CRediT authorship contribution statement

Omur Gulsum Deniz: Visualization, Methodology. **Suleyman Kaplan:** Visualization, Supervision, Methodology, Writing – review & editing. **Hamza Mohamed:** Writing – review & editing, Writing – original draft, Methodology.

Declaration of Competing interest

The authors have no conflicts of interest to declare.

Data availability

Data will be made available on request.

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